

8-compartment model

Lumped-Parameter Model of the Cardiovascular System (8-Compartment)

Conceptual Framework

The circulation is a closed loop of compartments, each one defined by two relationships: a **pressure-volume relationship** (how the compartment's pressure depends on its volume) and a **flow relationship** (how pressure gradients between two compartments determine the blood flow between them). For vascular compartments these relationships are simple and static. For cardiac chambers the P-V relationship is time-varying, reflecting the cardiac cycle of contraction and relaxation.

Circuit Topology

Eight compartments form a single closed loop. Blood flows:

$$\text{LA} \xrightarrow{R_{mv}} \text{LV} \xrightarrow{R_{av}} \text{Sys Art} \xrightarrow{R_{as}} \text{Sys Vein} \xrightarrow{R_{vs}} \text{RA} \xrightarrow{R_{tv}} \text{RV} \xrightarrow{R_{pv}} \text{Pulm Art} \xrightarrow{R_{ap}} \text{Pulm Vein} \xrightarrow{R_{vp}} \text{LA}$$

Each connection between compartments behaves like a fluidic Ohm's law resistor: $Q = \Delta P / R$. The four valve connections (mitral, aortic, tricuspid, pulmonic) additionally enforce unidirectional flow via a diode function.

Pressure-Volume Relationships

Vocabulary

Compliance of a compartment: $\Delta V / \Delta P$

Elastance of a compartment: $\frac{1}{C} = \frac{\Delta P}{\Delta V}$

Stiffness is the instantaneous version of elastance: $\frac{\delta P}{\delta V}$, often used interchangeably with elastance.

The subscript labeling scheme is artery/vein (a/v) followed by circulation type, namely systemic or pulmonary (s/p). So P_{ap} is the pressure in the pulmonary artery; V_{vs} is the volume in the systemic vein, etc.

Vascular Compartments (static)

For each of the 4 major vascular compartments $j \in as, vs, ap, vp$:

$$P_j = V_j / C_j$$

where C_j is a fixed compliance.

Cardiac Chambers (time-varying)

For each cardiac chamber $i \in LV, RV, LA, RA$, pressure depends on both volume and time because the myocardium actively contracts and relaxes over the course of the cardiac cycle. At each moment in time there is a distinct P-V relationship - starting at the fully relaxed state (end of diastole), progressing to the fully contracted state (end of systole), and then back to the relaxed state. The convention is to explicitly define two P-V relationships, namely a linear **end-systolic** relation and an exponential **end-diastolic** relation. The P-V relationship at any other time in the cardiac cycle is obtained by interpolating between these two limiting curves using a periodic activation function $\epsilon_i(t)$ that cycles between 0 and 1:

$$P_i(t, V_i) = \epsilon_i(t)P_{es,i}(V_i) + (1 - \epsilon_i(t))P_{ed,i}(V_i)$$

Recognizing that each chamber has its own P-V parameters we can drop the " i " for readability.

End-systolic P-V relation (ESPVR):

$$P_{es}(V) = E_{max}(V - V_0)$$

End-diastolic P-V relation (EDPVR):

$$P_{ed}(V) = A(e^{B(V-V_0)} - 1)$$

Note that all compartments have the property that there is a small amount of blood that can be added to the empty chamber before any increase in pressure occurs - this is called the unstressed volume, V_0 . In the vascular compartments we ignore it and only refer to stressed volume. But for several reasons it's common to be explicit about the unstressed volume of a cardiac chamber; that said, when solving the inverse problem we typically ignore these volumes (set them to a constant).

$E_{ed,ref}$ reparameterization. Solving the inverse problem for A is challenging because it has an enormous, poorly conditioned dynamic range. So we reparameterize the EDPVR in terms of its derivative, termed E_{ed} , at a pre-specified reference volume, V_{ref} :

$$E_{ed,ref} = A \cdot B \cdot e^{B(V_{ref}-V_0)}$$

Rather than treat (A, B) as free parameters of the EDPVR, we instead solve for $(E_{ed,ref}, B)$. V_{ref} is a fixed CONSTANT, set near a typical end-diastolic volume (EDV) for each chamber (e.g. $V_{ref,RV} \approx 125$ mL, $V_{ref,LV} \approx 120$ mL, $V_{ref,LA} \approx 35$ mL, $V_{ref,RA} \approx 30$ mL).

When solving the forward problem (simulation) we still need to be able to compute the EDPVR, so it's convenient to keep the A -parameterization handy:

$$A = \frac{E_{ed,ref}}{B \cdot e^{B(V_{ref}-V_0)}}$$

After the simulation converges and the actual EDV is known (which may differ from V_{ref}), we can compute the stiffness at that EDV. This is

known as the "operating-point stiffness", $E_{ed} = AB e^{B(EDV - V_0)}$, which is considered a valuable diastolic property of the chamber.

Activation function $\epsilon(t)$: a dimensionless "timer" that rises from 0 (full relaxation) to 1 (full contraction) and decays back to 0 over the course of one cardiac cycle, of duration $1/HR$ seconds (HR = heart rate). $t = 0$ corresponds to a chamber's *onset* of systole, or equivalently, its *end* of diastole (fully relaxed). The duration of systole is T_{es} and the relaxation time constant is τ :

$$\epsilon(t) = \begin{cases} \frac{1}{2} \left[\sin \left(\frac{\pi}{T_{es}} t - \frac{\pi}{2} \right) + 1 \right] & t < \frac{3}{2} T_{es} \\ \frac{1}{2} e^{-(t - \frac{3}{2} T_{es})/\tau} & t \geq \frac{3}{2} T_{es} \end{cases}, \text{ for } t \in [0, 1/HR]$$

The ventricles share timing parameters ($T_{es,v}, \tau_v$) with systole beginning at $t = 0$ (onset of ventricular contraction). Since the atria contract first, followed by the ventricles, their timing is out of phase. The atria share timing parameters ($T_{es,a}, \tau_a$) with systole beginning at an offset of AVD (atrio-ventricular delay) milliseconds before ventricular systole. So the atrial local time is $t' = (t - AVD) \bmod (1/HR)$. This offset places the atrial kick in late ventricular diastole.

Flow Relationships

All flow between adjacent compartments follows Ohm's law:

$Q_{j \rightarrow k} = (P_j - P_k)/R_{jk}$. Valve connections additionally enforce unidirectionality:

Vascular resistors:

$$Q = \frac{P_{upstream} - P_{downstream}}{R}$$

Valve resistors (resistance + diode):

$$Q = \frac{\Delta P}{R} \delta, \quad \delta = \begin{cases} 1 & \text{if } P_{upstream} > P_{downstream} \\ 0 & \text{otherwise} \end{cases}$$

The four valves and their upstream/downstream compartments:

Valve	Resistance	Upstream	Downstream
Mitral	R_{mv}	Pulm Vein (or LA)	LV
Aortic	R_{av}	LV	Sys Art
Tricuspid	R_{tv}	Sys Vein (or RA)	RV
Pulmonic	R_{pv}	RV	Pulm Art

Note: The venous-to-atrial connections (R_{vp} : pulm vein $\rightarrow$ LA; R_{vs} : sys vein $\rightarrow$ RA) are simple resistors without valves.

System of ODEs

The state vector is comprised of a volume and pressure variable for each of the 8 compartments. But from the P-V relationships described above, each $P(t)$ can be directly calculated from the $V(t)$, so only 8 volume states are needed. Each volume evolves as inflow minus outflow. With flows in and out of compartment i denoted by Q , we have:

$$\frac{dV_i}{dt} = Q_{in,i} - Q_{out,i}$$

Explicitly, traversing the loop:

$$\begin{aligned}
\frac{dV_{LA}}{dt} &= \frac{P_{vp} - P_{LA}}{R_{vp}} - \frac{P_{LA} - P_{LV}}{R_{mv}} \delta_{mv} \\
\frac{dV_{LV}}{dt} &= \frac{P_{LA} - P_{LV}}{R_{mv}} \delta_{mv} - \frac{P_{LV} - P_{as}}{R_{av}} \delta_{av} \\
\frac{dV_{as}}{dt} &= \frac{P_{LV} - P_{as}}{R_{av}} \delta_{av} - \frac{P_{as} - P_{vs}}{R_{as}} \\
\frac{dV_{vs}}{dt} &= \frac{P_{as} - P_{vs}}{R_{as}} - \frac{P_{vs} - P_{RA}}{R_{vs}} \\
\frac{dV_{RA}}{dt} &= \frac{P_{vs} - P_{RA}}{R_{vs}} - \frac{P_{RA} - P_{RV}}{R_{tv}} \delta_{tv} \\
\frac{dV_{RV}}{dt} &= \frac{P_{RA} - P_{RV}}{R_{tv}} \delta_{tv} - \frac{P_{RV} - P_{ap}}{R_{pv}} \delta_{pv} \\
\frac{dV_{ap}}{dt} &= \frac{P_{RV} - P_{ap}}{R_{pv}} \delta_{pv} - \frac{P_{ap} - P_{vp}}{R_{ap}} \\
\frac{dV_{vp}}{dt} &= \frac{P_{ap} - P_{vp}}{R_{ap}} - \frac{P_{vp} - P_{LA}}{R_{vp}}
\end{aligned}$$

Volume conservation: the sum of all compartment volumes is constant, equal to the total “stressed volume”, V_S :

$$V_S = V_{as} + V_{vs} + V_{ap} + V_{vp} + (V_{LV} - V_{0,LV}) + (V_{RV} - V_{0,RV}) + (V_{LA} - V_{0,LA}) + (V_{RA} - V_{0,RA})$$

We let V_S be a parameter of the system (ie known), leading to independence among 7 out of the 8 ODEs; one compartment volume (eg V_{vs}) can be calculated from the conservation constraint. V_S is one of the parameters identified when solving the inverse problem.

Parameters

Vascular (8):

Parameter	Description	Units
C_{as}	Systemic arterial compliance	mL/mmHg

Parameter	Description	Units
C_{vs}	Systemic venous compliance	mL/mmHg
C_{ap}	Pulmonary arterial compliance	mL/mmHg
C_{vp}	Pulmonary venous compliance	mL/mmHg
R_{as}	Systemic vascular resistance (SVR)	WU
R_{ap}	Pulmonary vascular resistance (PVR)	WU
R_{vs}	Systemic venous $\rightarrow$ RA resistance	WU
R_{vp}	Pulmonary venous $\rightarrow$ LA resistance	WU

Valve (4):

Parameter	Description	Units
R_{mv}	Mitral valve resistance	WU
R_{av}	Aortic valve resistance	WU
R_{tv}	Tricuspid valve resistance	WU
R_{pv}	Pulmonic valve resistance	WU

Cardiac (per chamber, $\times 4 = 16$): $i \in LV, RV, LA, RA$

Parameter	Description	Units
$E_{max,i}$	End-systolic elastance (contractility)	mmHg/mL
$E_{ed,ref,i}$	End-diastolic elastance at $V_{ref,i}$	mmHg/mL
B_i	EDPVR exponential curvature (intrinsic stiffness)	1/mL
$V_{0,i}$	Unstressed volume	mL

Timing (5):

Parameter	Description	Units
T_{es}	Ventricular systolic duration	ms
τ	Ventricular relaxation time constant	ms
$T_{es,a}$	Atrial systolic duration	ms
τ_a	Atrial relaxation time constant	ms
AVD	Time between atrial systole and ventricular systole	ms

Global (2):

Parameter	Description	Units
V_S	Total stressed blood volume	mL
HR	Heart rate	bpm

Fixed constants: $V_{ref,i}$ for each chamber.

Total number of tunable parameters: 35 (12 vascular/valve + 16 cardiac + 5 timing + 2 global). In practice we do not identify all 35, but rather fix a subset of these, eg the valve resistances, vein-to-atrial resistance, and cardiac chamber unstressed volume V_0 . HR is measured rather than estimated. So in total we might identify 24 free parameters.

Boundary Conditions

All states satisfy periodic boundary conditions: $V_i(0) = V_i(1/HR)$. In practice, we formulate the system of ODEs as an initial value problem and integrate over >5 cardiac cycles, adaptively checking for convergence to a periodic orbit.